

Using LLM Digital Twin Simulation to Evaluate Synchronous Group Interaction: With an Application to Focus Group Design*

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Abstract

We use digital twin simulation to model group dynamics under different synchronous discursive designs, at scale and across repeated trials – revealing the sampling distribution of the qualitative results under each design. These methods generalize to any synchronous discussion design, and enable us to state the assumptions required for the simulation to add to the accumulation of knowledge as well as validation methods that can help weaken these assumptions. As an application, we use these simulation methods to understand the properties of focus group research design choices. Our results show that in repeated trials, the traditional, small-N synchronous design either neglects under-represented viewpoints, or it generates text data that is uncorrelated across trials, indicating a “garden of forking paths” discursive dynamic. In contrast, we evaluate a novel large-N *structured group design* under which text data remains stationary across repeated trials and responses, and captures diverse respondent perspectives and reasoning.

Keywords: Digital Twins Simulation, Large Language Models, Deliberation, Algorithmic Curation, Synchronous Design, Focus Groups

1 Introduction

Human opinions are formed through social interactions (Chambers, 2018; Habermas, 1984). Learning how public opinion emerges from social interactions requires synchronous research designs that collect closed- and open-ended data while participants actively engage each other (Esterling et al., 2021).¹ The design or context for those interactions, however, heavily conditions the nature and dynamics of discussion (Dryzek et al., 2019; Esterling, 2018; Fishkin, 2009; Niemeyer et al., 2024), and so there is a need to understand the *properties* of alternative discussion designs for evaluating emergent group interaction.

An ideal method for evaluating the properties of discussion designs would require hosting a discussion and collecting data using a specific experimental design many times over, across different designs, each time with human participants. This procedure would reveal the sampling distribution of results generated under each design. Using human participants in synchronous designs, however, is extremely expensive and time consuming – particularly when the design scales to require hundreds or thousands of participants that must meet at the same place and time.² Here we develop low-cost methods for digital twins LLM simulation to explore the emergent properties of group discussion under different discursive contexts and designs, and at arbitrary scale (Wang et al., 2025; Zhang et al., 2024). We specify validation tests and state the assumptions required for digital twins simulation methods to contribute to the accumulation of knowledge for discursive design.

The digital twins methods we introduce are general to any group discursive setting. In this paper, we specifically apply the methods to investigate the properties of alternative *focus group* research designs. The focus group is a core method in qualitative and

¹Asynchronous platforms cannot replicate the dynamics of human discursive interaction as they tend to reduce spontaneity, shorten answers, and create uneven discussion flow and unfocused exchanges compared to synchronous platforms (Zhang et al., 2024).

²To give a sense of the costs, the Connecting to Congress town hall experiments (Neblo et al., 2018) used probability sampling to recruit participants for online synchronous meetings, and each session of 20-30 participants cost about \$10K to host. The Our Budget, Our Economy synchronous in-person deliberation event (Esterling et al., 2021) recruited over 2,000 participants, and the project required a year of planning and a budget of over \$2M.

mixed-methods academic research, and also is widely applied in marketing and political consulting. The focus group brings together individuals to discuss a set of questions in a moderated group discussion, providing qualitative insights not only into what opinions, attitudes and beliefs participants hold, but more importantly, *why* they hold those opinions and *how* those opinions emerge from interpersonal interactions (Cyr, 2019).

The traditional focus group design relies on small group discussions, typically of six to eight participants to ensure that all participants have a voice. The small group design, however, is vulnerable to undesirable group dynamics, including *unrepresentativeness*, *path dependence* based on the specific individuals present and the speaking order, and *group think* where the conversation is dominated by a given perspective. These vulnerabilities limit the ability of focus group results to generalize inferentially to a larger population and to produce meaningful or valid discursive data.

Here we propose a redesign of the focus group to enable scaling to a large-N research design, which in turn will address these limitations. We introduce the *structured group design* that, in contrast to the traditional focus group design, can scale to any size group. The structured group design enables synchronous participant input at scale through a three-stage discussion format centered on a substantive topic, motivated by a closed-ended survey question, relying on an LLM pipeline to capture diverse perspectives (similar to Lu et al., 2024). Our project contributes to an emerging literature on using LLMs to develop deliberative systems and platforms (Estornell and Liu, 2024; Karacapilidis et al., 2024; Ma et al., 2024; Tessler et al., 2024; Zhang et al., 2024).

We then use digital twins LLM simulation methods to explore the properties and behaviors of the structured group under both the small-N and the large-N design. The simulation shows that the small-N design is vulnerable to at least two pathologies. First, the small-N design often neglects underrepresented viewpoints, and second, among the viewpoints discussed, the qualitative text data collected is poorly correlated across replications, indicating that conversations in a small group follow an idiosyncratic “garden of

forking paths” logic where the focus group conversation does not systematically capture opinions within a population. In contrast, the large-N design robustly surfaces even underrepresented viewpoints and explores the full conversation space that exists in a larger population. We identify the assumptions required in order for the digital twins simulation to contribute toward the accumulation of knowledge regarding discursive designs, and we show validation methods that can help to weaken those assumptions.

2 Limitations of the Traditional Focus Group Design

The focus group is a method of qualitative and mixed methods research that brings together individuals to discuss a set of open-ended questions that a researcher wishes to investigate. Usually, the focus group will have between six and eight participants to ensure that each can contribute to the discussion. A trained facilitator will guide the discussion, allowing participants to use their own words, categorizations, concepts and associations (Parker and Tritter, 2006). Unlike survey research, the participant responses are not independent; in a focus group the revealed attitudes and beliefs are generative from the social interactions (Kitzinger, 1994; Smithson, 2000). The resulting data is a qualitative corpus that can be analyzed for individual beliefs, group-level reasoning, and social interactions that unfold during the meeting (Cyr, 2019; Parker and Tritter, 2006).

In principle, the focus group design enables the researcher to collect more nuanced data on opinions than can be captured in a standard survey. The group-based discussion replicates the kinds of everyday social interactions that generate opinions and can reveal meanings and understandings from the participant’s own (emic) perspective (Cyr, 2019; Habermas, 1984), revealing not just what opinions and beliefs individuals hold but why they hold them (Chambers, 2018). Engaging in group-based interaction can lead participants to engage in deeper cognitive processes (Tetlock, 1983) rather than just offering top of the head responses like in a standard survey (Zaller and Feldman, 1992). Focus

groups enable the researcher to explore new topics in opinion research that can inform and refine survey measurements. And focus groups can be designed to create the kind of social support that facilitate the discussion of sensitive topics (Woodyatt et al., 2016).

In the traditional focus group design, the number of participants must remain small – generally between six and eight participants per session – so that all participants have an opportunity to speak. The small-N design creates vulnerabilities, however, that make the results likely not generalizable or representative of an emic, interpersonal understanding among members of a population for the issue under discussion. As a practical matter, it is not possible for any such small group of individuals to be representative of the diverse perspectives in a larger population, even if the selection is by random or stratified sampling (Boas, 2024; Conover and Searing, 2005; Leung and Savithiri, 2009).

Moreover, small group dynamics make the discussion within the focus group heavily dependent on the specific individuals present (Parker and Tritter, 2006). If the individuals who happen to be present are not sufficiently diverse, it is possible that some perspectives in the population are never represented or expressed in the discussion – and a small-N inherently cannot encompass the perspectives of diverse populations. In addition, discussion within a small group where each person speaks in turn is likely to create a path dependence based on who is talking and the order in which they speak (Cyr, 2019). The conversation will navigate idiosyncratically down one path within a garden of forking paths and not engage in a full exploration of the conversation space (Neblo et al., 2019). Finally, small groups are vulnerable to “group think” where the conversation is dominated by one specific perspective (Parker and Tritter, 2006; Sunstein, 2002).

We posit that discovering the systematic discussion on a topic or issue within a population, or that would occur within a population, is the goal of focus group research.³ That is, the focus group should create a data generating process where the discussion data that results from the focus group matches or reflects the discussion that would occur

³In much of the focus group methods literature, authors assert that focus group research is only useful for exploratory research. Our methods are able to extend focus group methods to inferential research.

within a larger population, had the larger population engaged in a similar discursive effort (Fishkin, 2018). Unfortunately, the traditional design of the focus group is unlikely to yield data that recovers this population-level perspective. The unrepresentativeness of participants coupled with path dependence and group think limit exploration of the full discussion space. A preferred design will ensure diverse viewpoints are expressed, and ensure that discussion is not path dependent and will not be vulnerable to group think.

3 The Structured Group Design

We propose an LLM-assisted focus group design for a synchronous meeting that can scale to any N, which we call the *structured group* design. The user interface design and algorithmic curation pipeline to implement the structured group design has been implemented in the collabinar platform called **Prytaneum**, which is built by the **TeCD-Lab** at UCR – a platform that is open source and free to use.⁴ Scaling the focus group to a large N addresses many of the limitations with the traditional focus group. In addition, the structured design, by design, reduces the risk of group think within a discussion, even in the case when the group remains small.

The structured group design involves three stages, for each question of interest. In the first stage, the facilitator first provides background information and framing, the ground rules and motivation, and then presents a question to the audience with a set of closed-ended response options paired with a text box requiring participants to explain the reasoning why they chose their preferred option. In our primary simulation below, we use the question, What is your preferred approach to developing nuclear power? With response options 1: “Implement rapid expansion and investment of nuclear power;” 2: “Maintain the current nuclear energy operations without change;” 3: “Phase out existing

⁴This design can be implemented as an asynchronous survey using three waves. However, the asynchronous survey design would lack a facilitator and discussion lead, and would likely experience significant attrition across the three waves, and suffer from the limitations of asynchronous platforms such as reduced spontaneity, shorter answers, and uneven discussion flow and unfocused exchanges compared to synchronous platforms (Zhang et al., 2024).

nuclear plants and halt new construction;” 4: “Prioritize researching improved implementations of nuclear power.” Once the facilitator closes the poll, the collabinar algorithm will generate a summary of the participants’ reasoning to support each of the four options. Of course, if none of the respondents chooses a given option there will be nothing to summarize and the summary for that option will be empty.

In the second stage, the facilitator will share the summarized perspectives with the focus group participants, and then ask the participants to read a reason supporting one of the options they did not initially choose (among the options that have a non-empty summary), and then to write a counterargument to that reasoning.⁵ Once the second stage ends the algorithm will summarize the counterarguments submitted for each of the non-empty options. Then in the third stage, the facilitators will ask participants to read a counterargument to the preferred option they selected in stage 1 and then ask them to write a response to the counterargument. The summary of the responses to the third stage are then summarized by the algorithm.

The structured group design scales to any number of participants. Irrespective of the number of other participants, each participant has a robust opportunity to express their views regarding the question under discussion. Assuming a large number of participants, and provided the options are well-constructed, there should be sufficient reasoning to support each of the options to reflect the full distribution of perspectives in the underlying population of interest from which the participants are drawn, analogous to the concept of inclusive discursive representation in deliberation theory (Dryzek, 1990). Importantly, since each participant is asked to consider and respond to alternative perspectives, the design reduces the risk of group think (Sunstein, 2000). Further, with a large N, the discussion summaries should inferentially capture systematic features of the underlying discussion, and hence should be stable over repeated experiments. And finally, the full diversity and nuances of individual perspectives are captured in the responses across three

⁵Alternatively, the researcher can randomly assign an opposing viewpoint in the second stage. A complete randomization ensures that each non-empty option will receive counterarguments.

stages of discussion that are stored in the database and can be mined for further insights.

In contrast, small-N groups that follow this structured protocol can yield final summaries that are inconsistent across experiments or exclude any reasoning to support the options supported by underrepresented viewpoints. And if the small-N design shows these limitations under the structured group design, there is no reason to believe that switching to an unstructured traditional focus group design would remedy those limitations, and indeed would simply re-expose the discussion to the risk of group think.

4 Simulating Opinion Data: Digital Twin Prompting

We propose a general method to simulate the results of synchronous group interaction in order to understand the properties of alternative research designs. Specifically, we use LLM prompting to simulate the discursive data generation process using alternative versions of the structured group design. These simulations allow us to explore the properties of the different sample sizes, under differing sampling methods and response options, in repeated trials, at considerably lower cost and more efficiently than if we recruited humans to generate the data.⁶ To improve the simulation methods, we make use of a digital twins approach. Digital twins technology uses data from the physical world in order to anchor the simulation to the physical process being modeled, provided the simulation undergoes sufficient validation (National Academy of Sciences, 2024).

There is considerable work on digital twin silicon sampling for survey research (Arora et al., 2025; Brand et al., 2023; Chuang et al., 2024; Dillion et al., 2023; Hämäläinen et al., 2023; Korst et al., 2025; Sarstedt et al., 2024), along with several projects that curate databases to power a digital twins pipeline (Hu et al., 2025; Park et al., 2024; Toubia et al., 2025). To create personas, we pipe into the LLM prompt digital twin data regarding the persona the LLM is instructed to adopt, as well as background information and the previous stages of the discussion as context which improves consistency of LLM

⁶The simulations in this paper cost about \$20 total and require about two days to run.

performance (Zhang et al., 2024).

As with all LLM-based technologies, digital twin methods must be carefully validated before they can be relied on to add to the accumulation of knowledge (Bail, 2024; Bisbee et al., 2024; Estornell and Liu, 2024; Messeri and Crockett, 2024; National Academy of Sciences, 2024). In addition to hallucinations, LLMs are known to exhibit systemic biases inherited from the training data (Blodgett et al., 2020; Borkan et al., 2019; Caliskan et al., 2017; Feng et al., 2023; Hofmann et al., 2024; Hu et al., 2024; Motoki et al., 2024; Sorensen et al., 2024; Taubenfeld et al., 2024; Tjumatja et al., 2024) or from alignment processes (Santurkar et al., 2023). Digital twins and silicon sampling methods can introduce other distortions. For example, when prompted to respond from the perspective of a specific persona, LLMs can flatten or misrepresent the perspectives of identity groups because demographic covariates are not associated with text in the training data (Sarstedt et al., 2024; Wang et al., 2025). LLMs also perform less well in representing non-English speaking populations (Qu and Wang, 2024).⁷

To validate the simulations for our proposed structured group design, we evaluate the output of the ensemble of LLM personas to ensure that the LLMs are correctly responding to the prompt instructions – that the LLM is adopting the instructed persona and responding appropriately to the initial survey question and to the other LLM inputs. We also evaluate the summaries generated at the conclusion of each stage to ensure that they correctly capture the diverse reasoning in a systematic and unbiased way.

To evaluate the LLM persona output, we make use of three tests, relying on the proposed framework for “algorithmic fidelity” proposed by Argyle and her collaborators (2023). The first test is *backward continuity*, which requires that the LLM responses are consistent with the digital twin persona. To test backward continuity, we mask the digital twin persona instructions that were included in the prompt and ask a research assistant

⁷In principle, LLM output can be debiased using regularization, fine-tuning, or alignment processes such as reinforcement learning from human feedback (Agiza et al., 2024; Bang et al., 2021; Bolukbasi et al., 2016; Dixon et al., 2018; Jin et al., 2021; Lin et al., 2024; Liu et al., 2020, 2021). Debiasing would be difficult or impossible to implement with human participants.

to infer the covariate values from the LLM generated text. The second test is *frontward continuity*, which tests whether the LLM output is a sensible response to the prompt. To test for frontward continuity, we ask a research assistant to read the prompt and assess whether the generated text is a natural or sensible response to the prompt.⁸ The third test is *coverage*, which is to compare the generated responses among the LLMs to a known list of arguments that have been generated by humans on the corresponding topic, and evaluate whether the LLMs express the full diversity of human viewpoints on the topic.

We also assess validity, bias, and quality of the abstractive summarization methods (Altemeyer et al., 2025; Mittelstadt et al., 2023; Zhang et al., 2020). To conduct this test, we use cosine similarity to measure closeness of each LLM response to the corresponding summary and evaluate whether distances from the summary are random or systematically related to the digital twin demographic covariate values (Wang et al., 2025).

5 Simulation Methods

Here we make use of the digital twins survey database provided by Toubia et al. (2025). In the simulations, we sample from the 2,058 Toubia survey respondents and use the covariate information to create a unique digital persona via prompt engineering, requesting the LLM to adopt the persona reflected in the digital twin covariates in order to provide open- and closed-ended survey responses. Since the respondents in the digital twin database are representative of the U.S population, random samples from the database provide covariate information that reflects the joint distribution in the U.S. population.

We created simulated structured group data by using a digital twins approach to design an ensemble of LLM personas – that is, a set of LLMs prompted to have diverse views driven by covariates that reflect the U.S. population. Using the structured (multi-round) group protocol described above, we have the LLMs engage repeatedly in trials

⁸Frontward continuity also stands as a Turing test (Aher et al., 2022), which requires that the text LLMs generate are indistinguishable from human generated text.

under alternative design options, with different topics and survey prompts. We calculated pairwise cosine similarity comparisons for the resulting text generated within each of the response categories across repeated simulated structured group trials under each design. To calculate the cosine similarities, we first convert the text output to embedding vectors which represent the semantic meaning of the relevant comparison text.

The cosine similarity between two trials evaluates the stability of response summaries across these repeated trials. If a synchronous design inferentially explores the full conversation space for a topic, then the summaries across trials will be semantically similar resulting in high cosine similarity scores on average with low variation across trials. If the design results in idiosyncratic discussions then the summaries across trials will be dissimilar, resulting in low cosine similarity scores across trials and high variance.

We use an LLM prompt designed for each of the three stages of a structured group trial, as well as a prompt to generate summaries of the reasoning offered at each stage. Our core experiments focus on the question listed above regarding nuclear power.⁹ For each prompt we specify parameters to determine the “persona” of the LLM participant using digital twin covariates that explain variation in attitudes toward nuclear power, along with an additional parameter that decides which LLM to use for that persona.¹⁰ The covariates are *education-level* (less than college degree, college or more), *gender* (male, female), *political ideology* (liberal, conservative), and *environmentalism* (pro-environmental movement, not pro-environmental movement). The prompt for stage 1 is:

You will adopt the personality of a [education-level]-educated, [gender], [ideology]-leaning participant of a focus group discussing the adoption of nuclear power. You have [level of affinity] for environmentalism. You must select from one of the four possible standpoints on nuclear power and briefly provide your

⁹We use the nuclear power question in our core experiments since nuclear power is not a polarizing topic. Below we also explore designs using the more partisan the topics of gun control and expanding the U.S. House of Representatives.

¹⁰We diversify the LLMs so the results are not confined to a specific LLM. We used 4 LLMs: Gemini 3.0 flash preview, Qwen3 32B, Llama 3.3 70B versatile, and OpenAI GPT OSS 120B, with the latter three being open weight LLMs. Each persona used the same LLM model across the three rounds of the focus group.

reasoning for doing so in 1-2 sentences.¹¹ The possible standpoints are:

- 1: Implement rapid expansion and investment of nuclear power.
- 2: Maintain the current nuclear energy operations without change.
- 3: Phase out existing nuclear plants and halt new construction.
- 4: Prioritize researching improved implementations of nuclear power.

We then request Google Gemini to summarize the reasons offered for each standpoint option (among non-empty options) using the following prompt:

You are a helpful assistant tasked with summarizing argumentative statements on the topic of nuclear power. I have gathered a list of statements that participants in an online focus group platform have provided as their reasoning to [standpoint option text]. Summarize the following statements as no more than five concise summaries. Your summaries should be worded as passive tone viewpoints, so that they capture as many viewpoints presented in the given statements as possible.

The LLM’s responses in stage 1 create three inputs to the second stage: *stage1-standpoint* (the participant’s chosen standpoint from stage 1), *stage1-reasoning* (the participant’s reasoning for choosing stage1-standpoint), and *opposing-viewpoint* (a randomly selected summarized reason to support a standpoint that the persona did not choose in stage 1). The prompt for stage 2 is:

You will adopt the personality of a [education-level]-educated, [gender], [ideology]-leaning participant of a focus group discussing the adoption of nuclear power. You have [level of affinity] for environmentalism. You have previously stated that your standpoint is to [stage1-standpoint], and your reasoning for this was: ‘[stage1-reasoning]’. As part of the focus group discussion, a different viewpoint has been brought up: ‘[opposing-viewpoint]’. You must now provide a counterargument for this viewpoint on nuclear power in 1-2 sentences.

¹¹The prompt also provides background information as context to help the LLM model personalities. The background we use is taken from this PEW report: <https://www.pewresearch.org/short-reads/2025/10/16/support-for-expanding-nuclear-power-is-up-in-both-parties-since-2020/>. “To help you choose an option, please know that according to PEW Research Center, about half of US residents favor expanding nuclear power, while the other half prefer to maintain operations without change or to phase it out. Most men favor expanding it while most women prefer phasing it out or keeping it the same. Republicans are more likely to favor expanding compared Democrats. Environmentalists have mixed opinions, where some favor expanding it because nuclear has a low-carbon footprint, while others favor contracting it because they are concerned about the risks of nuclear waste disposal.”

The responses from stage 2 are summarized in a manner similar to the summarizations in stage 1, which creates one more input to the third stage, *counterargument* (a summarized response from stage 2 that argues against this participant’s original standpoint). The prompt for stage 3 is:

You will adopt the personality of a [education-level]-educated, [gender], [ideology]-leaning participant of a focus group discussing the adoption of nuclear power. You have [level of affinity] for environmentalism. You have previously stated that your standpoint is to [stage1-standpoint], and your reasoning for this was: ‘[stage1-reasoning]’. As part of the focus group discussion, a counterargument to your standpoint has been brought up by a participant who said that: ‘[counterargument]’. You must now respond to this counterargument in 1-2 sentences.

The responses in stage 3 are summarized again within each (non-empty) standpoint. The simulation then saves the participant-level responses for each stage along with all inputs such as the covariates defining the participant’s persona and opposing standpoint used in stage 2. It also saves the summaries from each round. For illustration, the appendix provides the summarized responses for the three stages, for a single N=60 group, showing the arguments raised in the structured group.

6 Results

Making use of this structured group design, we ran 50 trials of the small-N version with six participants each; 30 trials of the medium-N version with 60 participants each; and 20 trials of the large-N version with 600 participants each. Each “participant” was one of the randomly selected digital twin LLM personas. Then for each standpoint, we create an $N \times N$ matrix of pairwise cosine similarities, measuring the similarity of the summary of stage 3 responses for each trial with the summary of stage 3 responses for every other trial within the standpoint and within that design. That is, each cell in this matrix is a cosine similarity comparison of the stage three summary for run x with run y within the given standpoint and design. We retain only the lower triangle of this matrix, excluding

the diagonal. If a summary is empty for a given standpoint on a given trial, the cosine similarity with every other trial is zero by definition.¹² For the $N = 6$ participant version trials we have 1,125 pairwise comparisons; for the $N = 60$ trials we have 435 comparisons; for the $N = 600$ trials we have 190 comparisons.¹³

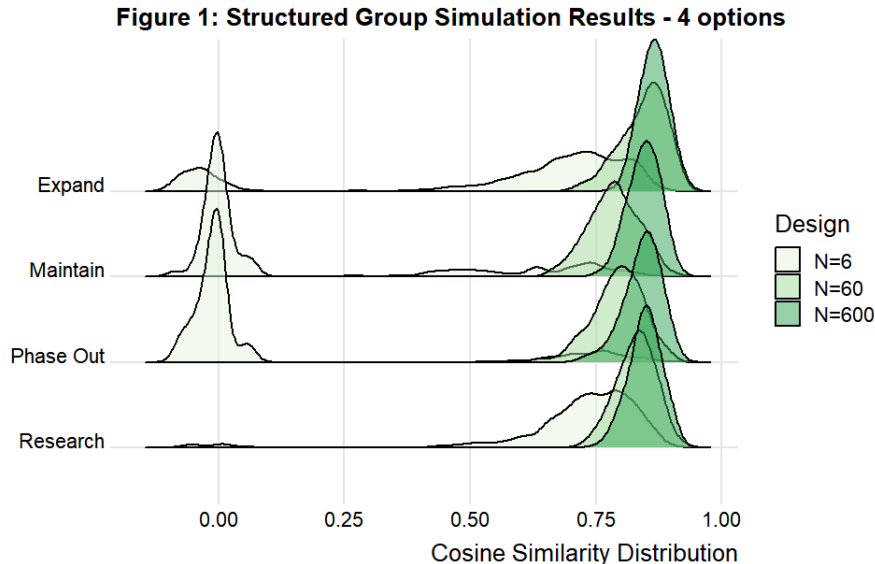


Figure 1: Distribution of pairwise cosine similarity scores, comparing between the small-N focus group design and the medium- and large-N design. Simulated focus group results on the topic of expanding nuclear power. Response 1: Implement rapid expansion and investment of nuclear power. Response 2: Maintain the current nuclear energy operations without change. Response 3: Phase out existing nuclear plants and halt new construction. Response 4: Prioritize researching improved implementations of nuclear power.

Comparing these similarity scores across trials helps to show the stability of responses within a given structured group design, within each standpoint option. Figure 1 compares the densities of the across-trial cosine similarities scores for designs with six, 60 and 600 participants, conditional on the response option. The within-response distributions of the cosine similarity scores show that the small-N version with size six have a mode at zero for each of the standpoints, indicating that the summary was empty for each standpoint with

¹²If the summaries are empty for two trials, we set the cosine similarity to zero.

¹³Each 600 participant simulation takes about an hour to run. Fortunately, since the results are nearly identical across trials, we do not need to run many trials to recover the large-N sampling distribution and 190 trials is a sufficient number.

some frequency. This mode at zero is small, however, for the fourth standpoint option which was to prefer researching improved implementations of nuclear power. This fourth option was the most popular among all LLM personas, but as a result, for focus groups with only six participants, that meant that for many trials there were no participants that favored one or more of the other options, and hence these other options were never even discussed in the focus group. In addition, when an option is discussed in a small-N group design, the cosine similarities indicate wide instability in the response across trials with non-zero modes centered at about 0.7 and wide dispersion – showing an interaction under the small-N design between the (in)frequency that an option is discussed and the randomness of the discussion.

By comparison, the 600 participant focus groups have a unimodal distribution with relatively low variance across trials centered at about 0.9. Conceptually, one can think of the sequence of experiments in the small-N design results in a multimodal distribution of divergent results while the sequence of experiments in the large-N design remains a unimodal stationary distribution. And importantly, the degree of instability of the small-N design itself varies across the response options, while the large-N design returns results that are always stable and of similar magnitude across the response options, irrespective of the relative popularity of each option. The performance of the trials with 60 participants is between these two extremes.

This simulation starkly shows the flaws in the traditional small-N focus group design, and the merits of the large-N design. Conceptually, just as with quantitative research, we see a strong instability of results across trials for the small-N design. This means that the results for the small-N design are not reliable representations of the underlying discussion that is the object of discovery in focus group research. Instead, small-N group dynamics create qualitative data that either reflects noise, or excludes underrepresented viewpoints in the population. In contrast, the large-N design yields a draw from a stationary distribution of outcomes that inferentially reflects a systematic summary of the discussion that

does or could occur in the population.

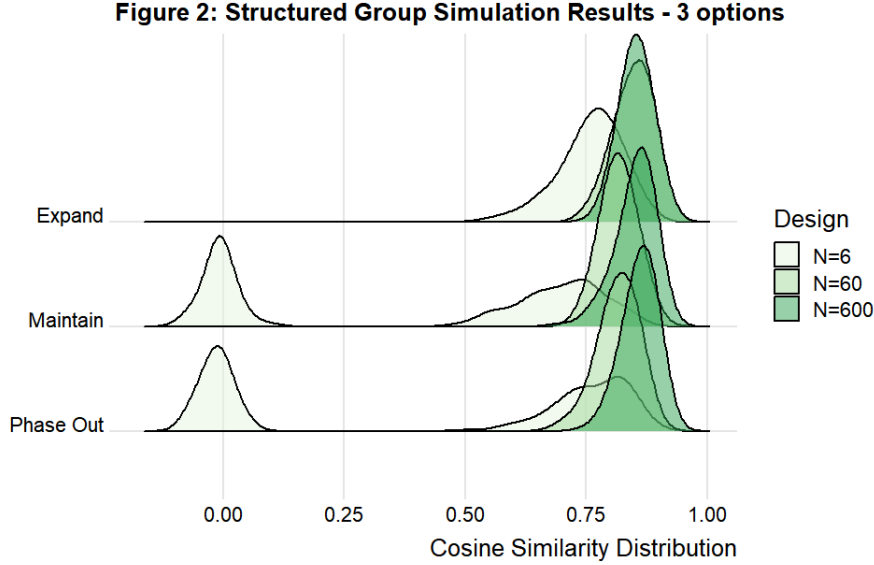


Figure 2: Distribution of pairwise cosine similarity scores, comparing between the small-N focus group design and the large-N design – three response options. Simulated focus group results on the topic of expanding nuclear power. Response 1: Implement rapid expansion and investment of nuclear power. Response 2: Maintain the current nuclear energy operations without change. Response 3: Phase out existing nuclear plants and halt new construction.

Notice that the “popular” standpoint option to research the implementation of nuclear power is unlike the other three options. That is, the other three options are to expand, to maintain the same level, or restrict nuclear power, which are regulatory and are mutually exclusive and exhaustive. All of the LLM personalities were attracted to the non-regulatory research option for whatever reason and hence the distributions of responses were skewed in our trials. We repeated the same sets of trials of six, 60 and 600 participants, but this time excluding the option to research nuclear power from the choice set which yields response distributions that are less skewed. The results of these trials are shown in figure 2. Note that when the response distribution is relatively unskewed, the performance of the 60 participant design nearly matches that of the 600 person design. This indicates that a larger sample size is required in the presence of underrepresented

viewpoints, and improved instrument design can reduce the N needed for reliable results. This simulation approach to understanding the sampling distribution of the data is analogous to recent proposals to use simulation for power analysis under complex research designs (Blair et al., 2019).

7 Stratified Sampling Illustration

Given that a small sample of six or eight participants will have a considerable amount of sampling variability, focus group researchers understand that stratified or even purposeful sampling can improve on simple random sampling to ensure diversity in a given group (Cyr, 2019). For example, researchers interested in methods of ideological depolarization will compose small groups that balance liberals and conservatives. We are able to use our digital twin simulation methods to explore benefits to small group research that can occur under such stratified sampling.

Of course, stratification does not help when covariates only weakly explain positions on topics such as nuclear power, but it might help improve the balanced representation of perspectives for topics where the stratifying covariates predict responses. Here we explore the consequences of stratification on self-reported ideology on the issue of gun control.¹⁴ We build the LLM personas using the same gender, education, ideology and model covariates we describe above, and run trials for $N = 6$ with simple random sampling, $N = 6$ where we stratify sampling to ensure that half of the participants are liberal and half are conservative, and $N = 60$ with simple random sampling.

We set three standpoint options for the LLM personas to choose from: “Gun control laws should be more strict,” “Gun control laws are about right,” and “Gun control laws should be less strict.” For context, we include the background information text: “To help you choose an option, please know that according to the PEW Research Center, 82

¹⁴For background, see the PEW report <https://www.pewresearch.org/short-reads/2024/07/24/key-facts-about-americans-and-guns/>

percent of U.S. conservatives believe that gun control laws should be the same or less strict while 92 percent of U.S. liberals favor making gun control laws more strict. That is, liberals favor making gun control laws more strict and conservatives to keep the laws the same or make them less strict.”

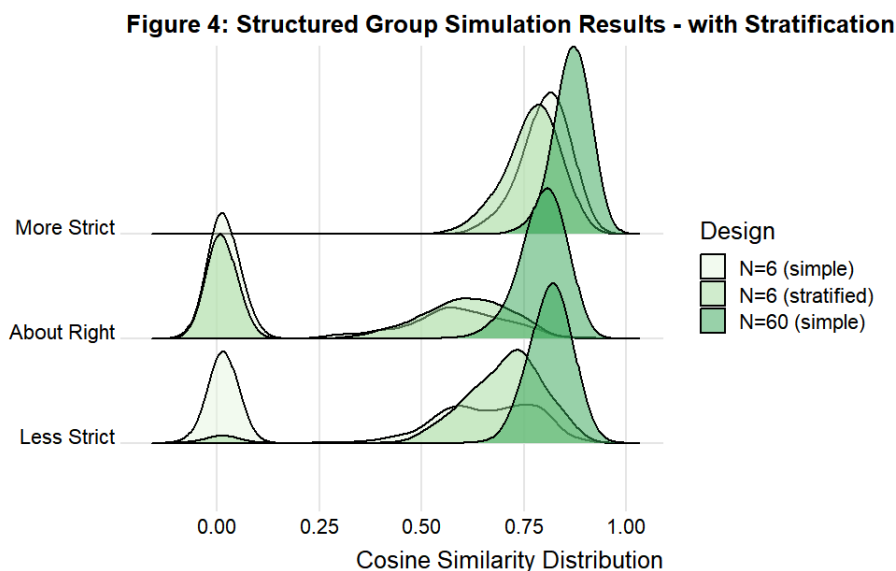


Figure 3: Distribution of pairwise cosine similarity scores, comparing between the small-N focus group design and the large-N design – three response options with stratified sampling comparison. Simulated focus group results on the topic of gun control. Response 1: Gun control laws should be more strict. Response 2: Gun control laws are about right. Response 3: Gun control laws should be less strict.

Figure 3 shows the results. Notice that stratification by ideology does help to ensure balance across the two polarized response options, to either be more restrictive or less restrictive, but it has no effect on the more neutral option to advocate the gun control laws are currently the right amount of regulation. The results show that this stratification design is effective if the purpose of the discussion is to ensure participants are speaking across differences, but less effective if the purpose of the discussion is to inferentially discover the full underlying discussion space.

8 Assumptions and Validation

The extent to which digital twin simulation can be relied on to add to the accumulation of knowledge depends on a set of application-specific assumptions (Bail, 2024; Bisbee et al., 2024; Estornell and Liu, 2024; Messeri and Crockett, 2024; National Academy of Sciences, 2024). While it is common for researchers to articulate assumptions for validity in statistical applications (Esterling et al., 2025), to date the LLM literature has not developed norms to state assumptions required for simulation research.

Our application of digital twins simulation to evaluating focus group design relies on two assumptions: 1) *Coverage*: with enough participants, the LLM-based discussion recovers the full range of human-made arguments within a given population on the topic of discussion across the three stages, and 2) *Unbiasedness*: that the discussion summaries systematically encompass the discussion space for the topic. Given these assumptions, we can infer the rate that underrepresented viewpoints are excluded from the discussion and the stability of the inferred discussion space across repeated trials. These are relatively weak assumptions that can be further weakened by our validation tests.

Importantly, note that there are two further, and very strong, assumptions that would be required in the ordinary case of survey research but that we do not need. These are 3) *Faithfulness*: that the digital twin personalities produce choices, reasoning or content that reflects the (intersectional) demographics of the digital twin covariates. As we discuss above, we know that the LLM training text data is not associated with demographic covariates, and hence LLMs are not well-designed to have the LLM output match the prompt demographic input (Wang et al., 2025). In our application, we use digital twins prompting to ensure that the arguments that LLMs voice in the focus group discussion ecologically capture the diversity of human argumentation, but our application does not require the argument content at the persona level to correspond to the demographic identity group’s voice reflected in the covariates.

We also do not require the assumption 4) *Distribution match*: that the digital twin

personalities select the standpoint options in a way that matches the marginal distribution of those preferences in the target human population. Our structured group design is motivated by the notion of discursive representation (Dryzek, 1990), and hence we only need to have a viewpoint expressed by at least one LLM participant for that perspective to be included in the corresponding summary. The marginal distribution of preferences among the participants is not relevant to discursive representation.¹⁵

We rely on validation tests to weaken our two assumptions. We use our backward continuity, frontward continuity and coverage validation tests to evaluate the coverage of human-made arguments in the discussion, and we rely on a bias audit to evaluate whether our summaries capture systematic perspectives.

To assess the LLM coverage of human arguments, we first consider the backward continuity test, where we masked the covariate information and asked a research assistant to infer the covariate values based on the generated text alone for 60 observations on the topic of nuclear power. This test checks if the LLM is following the prompt instructions to create a given persona. We rely on χ^2 tests from confusion matrices. We find that the research assistant is 26 percent more likely to correctly infer female rather than male ($p = 0.063$), 31 percent more likely to correctly infer college educated rather than not college educated ($p = 0.022$) and 25 percent more likely to correctly infer liberal rather than conservative ($p = 0.036$). In addition, the true environmentalism score is about a standard deviation higher when the research assistant inferred the persona was positive toward environmentalism ($p < 0.001$). Finally, we also masked which standpoint option the LLM chose, and the research assistant was able to infer the standpoint chosen from the text of the reasoning offered and made the correct inference 58 out of the 60 responses. Note that backward continuity weakens both the coverage and the faithfulness assumptions, although in our application we only need it to weaken the former.

¹⁵For example, if 100 LLM personas offer an identical perspective, and only one LLM persona made a different perspective, then the summary would include both perspectives. The marginal distribution of participants holding a given perspective is not relevant.

Second, we consider the frontward continuity test, where we asked the research assistant to read the prompt created for each persona and to evaluate if the LLM output was responsive to the text in the prompt. This is similar to a Turing test. Here the research assistant was asked to read all three rounds for a 60 participant trial. Out of the 180 opportunities for the LLM to be non-responsive, the research assistant found only five cases where the LLM did not respond to the prompt. However, when the lead author examined those five responses, he found the output to be responsive. By design, our pipeline does not depend on the memory of the LLM across the three rounds – instead, we pipe in the text from each previous round into the prompt. This method avoids the limitations of LLMs’ diminished performance in multi-turn conversations, and as a result the LLMs are responsive on each API call. One concern might be that LLMs are even more responsive to the prompt text than humans would be, creating a hyper responsiveness that would not be present in human-generated text.

Third, we implement validation tests for issue coverage. Here we make use of an existing reference listing of pro and con arguments regarding nuclear power from the *Encyclopedia Britannica*, which provides three pro arguments that nuclear power: 1) has zero emissions and can reliably and inexpensively support an electricity grid; 2) is a perfect complement to weather-dependent renewable energies; and 3) protects American global national security interests; along with three con arguments that nuclear power: 4) is dangerous, from the risk of meltdowns to the problem of nuclear waste; 5) is expensive to initiate and delays implementation of truly sustainable and renewable energy sources; and 6) is inextricably linked to lethal nuclear weapons.

We provided both a human coder and an LLM-based coder a candidate list of 60 persona reasons for and against nuclear power and asked each coder to decide if each candidate reason matches with an argument on the candidate list, and if so, which one. The two coders disagreed on 22 out of the 60 classifications. The lead author checked each instance of disagreement and found that seven disagreements were cases where the

candidate reason expressed both reference arguments (so both correct), the human coder made 11 errors and the LLM coder made 4 errors.

Using the LLM-coded classifications, in this session of 60 participants, the first four of the Britannica arguments were expressed in the first round of discussion but the last two con arguments, arguments 5 and 6, were not. However, argument 5, that investing in nuclear power diverts funding from renewable sources, was raised repeatedly in the counterarguments and responses to counterarguments in stages two and three of the discussion. And, it is reasonable to argue that the final con arguments, drawing the analogy to nuclear weapons, is a strange and not especially strong argument – so its absence from the discussion is not surprising. In addition, the LLM personas expressed eight additional arguments that are not found on this reference list. One can see all of the summarized arguments across the three stages for this trial in the appendix.

Finally, we develop a bias audit method to detect bias in the summarizations. Note that the stable distribution that results from the large-N design is normatively defensible only if the qualitative summaries of the responses are an unbiased reflection of the individual contributions made during the meeting. If the AI-based summary tool is biased, then the summaries that result would be objectionable in that they would systematically exclude certain perspectives. It is well-known that the underlying LLMs that drive AI applications can have biases inherited from their training data or from alignment or fine-tuning interventions (Borkan et al., 2019; Caliskan et al., 2017; Kambhatla et al., 2022; Liu et al., 2020; Santurkar et al., 2023). Thus it is important to conduct an audit of the LLM output to verify the results are unbiased.

We conduct this audit in two ways, first using the simulated data from our experiments which we report here, and then using a human-generated dataset which we report in the next section. To create the simulated audit data, we calculate the cosine similarity of each individual LLM-simulated persona’s reasoning with the summary of the standpoint option they chose. Recall that we ask the summary tool to include five different perspectives

within each summary. We calculate the cosine similarity between the simulated persona’s response and each of the five summarized perspectives, and then retain the maximum value. That is, if simulated individual i chose response option k , we calculated the cosine similarity between i ’s response and the five response k summarized perspectives, and then saved the maximum of these five values. For comparison to the human data we report below, we only use the text generated from the first stage of our experiments.

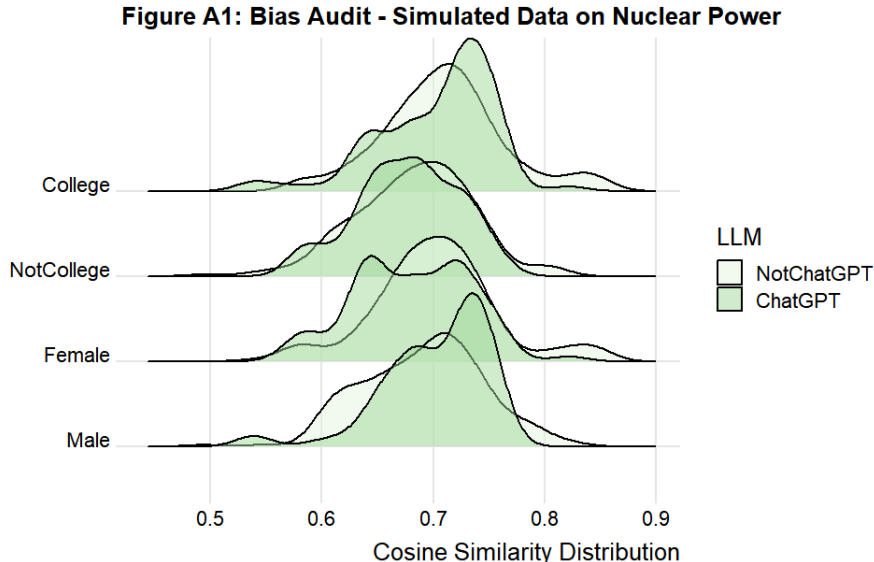


Figure 4: Distribution of simulated individual cosine similarity scores, conditional on demographics.

Figure 4 shows the conditional distributions of the cosine similarity scores for the simulated individuals, conditional on two covariates (education and gender) and the LLM (OSS-GPT versus the others). The conditional distributions indicate no systemic disparities by gender, or perhaps an under-representation of the female non-OSS-GPT perspectives. However, there is a slight over-representation of college-educated simulated individuals. As we show in the next section, these results replicate in the human data. Finally, there is a noticeable over-representation of OpenAI’s OSS-GPT, which is likely due to the fact that GPT provides more generic answers to the prompts.

Overall, in both the simulated and in the human-generated data, the conditional cosine

similarity distributions in the bias test of figure 4 are largely the same across all covariates. These results suggests that all diverse perspectives are well-captured and included in the summarized results. And of course, also in contrast to the traditional focus group design, our method allows for this kind of bias audit to detect systemic disparities, and the AI-based summarizer can be debiased so that the summaries are orthogonal to demographic characteristics in the event that they do fail an audit (Bolukbasi et al., 2016; Dixon et al., 2018; Liu et al., 2020). In a traditional focus group design, where individuals speak in turn, there is no method to detect or to correct these kinds of systemic biases that occur in real time during a free-flowing discussion.

9 Bias Audit with Human and Simulated Data

As we mention in the previous section, digital twin simulations have a limitation in that the personas that LLMs are prompted to adopt may not express their beliefs in a voice that matches the (intersectional) identity group they are intended to represent. The main reason for this possible limitation is that, while the LLM almost certainly has texts in its training data that were authored by members of the intersectional identity group, the LLM architecture does not connect those texts with demographic covariates. That is, the training data for the LLM likely has texts written by, say, a pro-environment, college educated, conservative female, but the LLM does not know which texts correspond to this demographic identity, and so when voicing this persona it can only make an inference about how such a person might express themselves. This limitation is only a concern for the bias audit, where we need to know the distance between a reason given and a corresponding summary for each participant given their demographic covariates.

To address this limitation, to further evaluate bias in the LLM summaries we make use of a human dataset that we have at hand which contains both text as well as covariates. The data come from a recent town hall held by the Institute for Democratic Engagement

and Accountability (IDEA) at the Ohio State University.¹⁶ The town hall discussion centered on topics related to the modernization of Congress, and the participants were a U.S. nationally-representative sample. After the town hall, the participants were asked to respond to a survey that collected demographic information and then they were asked to indicate their support or opposition to a proposal to expand the size of the U.S. House of Representatives, and they also were asked to provide their reasoning for their position. This data is very similar to the kind of data that would be collected in a structured group in that the responses with the reasoning that participants submitted followed from a town hall discussion where they were exposed to both pros and cons.

There were 586 responses to this question. We use our LLM summary tool to create summaries of the favor and oppose responses, then we calculate the cosine similarity of the individual’s reasoning with the summary of the position they chose. We use the same procedures as above to conduct this audit using human data. Specifically, we ask the summary tool to include five different perspectives within each summary. We calculate the cosine similarity between the individual’s response and each of the five perspectives, and then retain the maximum value. That is, if individual i was in favor of expanding the size of the House, we calculated the cosine similarity between their response and the five in-favor summarized perspectives, and then saved the maximum of these five values.

Figure 5 shows the distribution of these individual-level cosine similarity scores, conditional on demographic covariates and on support or opposition position. Note that the cosine similarity is higher for the support than the opposition position, which simply indicates that there are likely more diverse reasons to be in opposition than support. But conditional on position, the distributions do not show substantive, systemic disparities across the demographic categories. The view of college-educated individuals who are in favor of the reform might be slightly over-represented, which is not a surprise given that LLMs tend to be biased in favor of well educated responses (Santurkar et al., 2023). Cer-

¹⁶The data was collected under the Ohio State University IRB protocol 2019B0075

tain views of non-white respondents who are opposed to the reform seem to be slightly under-represented, given the mildly bimodal conditional distribution, which indicates that some of the perspectives in this group are not well-captured in the summaries.

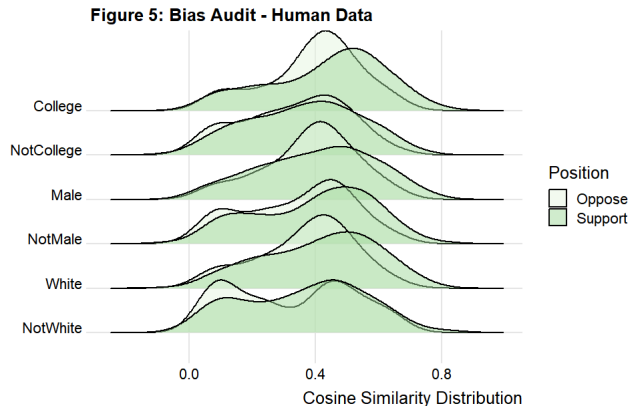


Figure 5: Distribution of human individual cosine similarity scores, conditional on demographics and position. Topic: Expansion of the U.S. House of Representatives

For comparison to the simulated results we report in the previous sections, we replicate the same focus group workflows that we report above, but this time using the topic of the expansion of the U.S. House of Representatives, and with only two response options: 1) increase the size of the U.S. House of Representatives by adding more seats, and 2) keep the size of the U.S. House of Representatives the same as it is now. To build the digital twin personas, we use the same pipeline and the same gender, education and ideology covariates as before; we remove the level of environmental movement support covariate; and add a race covariate, a need for cognition score, and a vertical collectivism score which denotes a focus on group loyalty combined with acceptance of hierarchical structures.¹⁷ As background information, we also include in the prompt “To help you choose an option, please know that expanding the size of the House will reduce the number of constituents in each district, enabling representatives to better connect with their district residents. However, adding new seats to the House would increase the cost of government and

¹⁷We add the need for cognition and vertical collectivism scores to the personas in order to ensure diverse responses to the prompt. In the absence of these covariates, liberal LLMs almost always chose to expand and conservative LLMs almost always chose to keep the House the same.

potentially make Congress more unwieldy.”

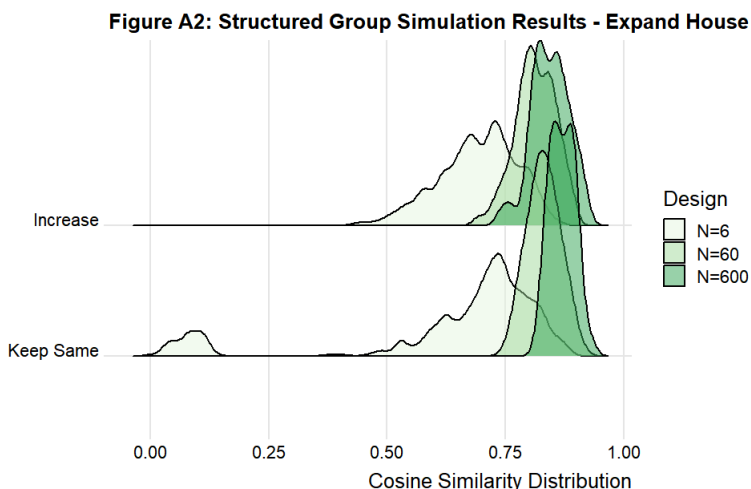


Figure 6: Distribution of pairwise cosine similarity scores, comparing between the small-N focus group design and the large-N design – two response options. Simulated focus group results on the topic of expanding the size of the U.S. House of Representatives. Response 1: In favor of expansion. Response 2: Opposed to expansion.

The figure 6 shows the distributions of cosine similarity scores across trials for each response. Notice that even with only two responses, there are many sessions where the small-N six person focus group never discusses the option to keep the House the same size. And as before, when a topic is discussed, the discussion shows considerable variability in the cosine similarity scores, indicating that each event produces relatively idiosyncratic discussion and data.

Figure 7 shows the bias audit results for the simulated data. Notice the strong similarities to the results of the bias audit between the data generated by humans as reported in figure 5 and in particular little evidence of bias in the summaries since the conditional distributions strongly overlap. The exception is that, as in the human data, college educated personas tended to have their responses more closely reflected in the summaries. Interestingly, even in the simulated data, supporters tend to have their arguments more closely reported in the summaries than opponents, which again suggests that there are more ways to oppose the policy than to support it. That the simulated data reflects

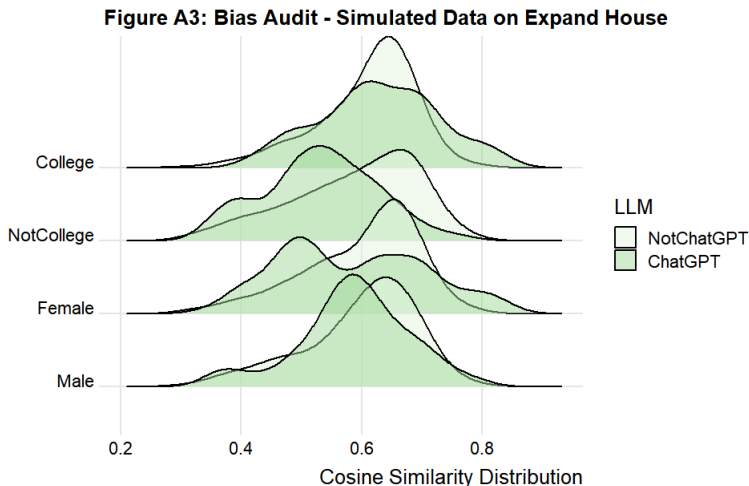


Figure 7: Distribution of simulated individual cosine similarity scores, conditional on demographics and position. N=600 design. Topic: Expansion of the U.S. House of Representatives

these patterns helps to reinforce that the digital twin simulated arguments have good correspondence with human authored arguments.

10 Demonstrating the Generality of the Methods

The methods we describe in this paper can apply to any synchronous research design. However, to assess a new design the Python script for the simulator would need to be rewritten. The author team collaborated to write the existing Python scripts by hand. However, to show the ease by which these methods can be applied to new designs, we used Claude Code to create a simulator for what we call the *unstructured group design*. The unstructured group design modifies the structured design by removing the closed-ended response options, and instead only asks the participants to brainstorm ideas on the topic through a multi-round discussion. The goal of this version was to show generality of the methods by asking Claude to modify the script to implement a different synchronous design. The new script for the unstructured design simulator only required two requests from Claude, an initial request and then a simple correction, and it took only minutes to

complete. The unstructured version worked first time off the shelf.¹⁸

11 Discussion and Conclusion

In this paper, we propose an LLM-based digital twins pipeline that can be used in any context to evaluate the properties of different synchronous designs. Digital twins simulation enables the researcher to explore the properties of alternative discussion designs computationally at low cost and effort by simulating the sampling distribution of qualitative data under the different designs. These methods enable the researcher to computationally understand the complex, emergent nature of human discussion under a given design, similar to how researchers use computational methods to understand the power of complex statistical models when an analytical solution is unavailable (Blair et al., 2019). These methods can be used to pilot alternative designs prior to investing in human trials. Finally, we state assumptions required for these simulations to add to the accumulation of knowledge for discussion designs, as well as validation tests that can weaken these assumptions.

As a proof of concept, we apply our simulation pipeline to evaluate alternative designs for focus group research. The application of our methods using the structured group design helps to show the properties of designs for human interaction, and in particular helps to show the sampling distribution of the qualitative data generated under alternative designs. In the main simulation, where the personas discussed the topic of nuclear power, we show the complex interaction between design elements from emergent behavior. In that case, a poorly designed survey instrument led to highly skewed data, which in turn led to results that either excluded perspectives, or resulted in data that was inherently noisy. In the replication simulation, we were able to compare outcomes between human and LLM-simulated debates, and the similarities in outcomes between simulated and human data helps to show the correspondence between the simulated world and the human world.

¹⁸The possibilities for alternative designs are endless. For example, an extension could be to use a stance extraction tool to assign opposite-stance reasons across the rounds to create discussion across differences for the unstructured group design.

Both of these applications use digital twins simulation to demonstrate the clear superiority of our proposed large-N structured group design to that of the traditional small-N design. In the traditional focus group design, a small and unrepresentative number of participants speak sequentially, a data generating process that creates path dependence and group think, and excludes underrepresented views. This results in qualitative data and inferences that lack generalizability. In contrast, the large-N focus group design enables a large number of participants to, in effect, speak simultaneously and to understand all of the diverse perspectives among participants, eliminating the problems of path dependence and bias in the data generating process. We also show that stratification can help for cases where responses are well-predicted by covariates.

Further, propose the coverage and unbiasedness assumptions needed to understand if the simulation can advance the accumulation of knowledge on research design, and validation tests designed to weaken those assumptions. In particular, we show that the LLM summary tool at the heart of our design is inclusive of diverse perspectives and does not seem to be vulnerable to systemic biases that can plague LLM-based tools.

The large-N structured group design is not vulnerable to the problems of unrepresentativeness, path dependence or group think that plagues the traditional small-N focus group design. Of course, the small-N design remains useful as a tool for exploratory qualitative research (Cyr, 2019), but there is no reason to limit the focus group methodology to exploration. Our proposed methods show how synchronous group methods can extend to inferential qualitative research, and can help researchers uncover a nuanced, interpersonal understanding of the merits of alternatives regarding a decision, in a way that meets the normative standards of deliberative theory (Dryzek et al., 2019; Niemeyer et al., 2024).

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A Appendix: Validation Data

To conduct the validation of the main simulation, we make use of a single trial for a focus group with $N = 60$. In this trial, the LLM personas debated the question of expanding nuclear power that omitted the non-regulatory option to continue to conduct research. Since each trial is a random draw from the LLM, it does not matter which trial we investigate – so we simply used the first trial as our data source for the validation. To find the validation data, navigate to the [reproduction GitHub repository](#) and you will find the data in the folder with trial id 4260266019. Table [A1](#) provides the text summaries for the three stages of the LLM persona discussion for this trial, where the first stage summarizes the initial reasons given, the second stage summarizes counterarguments to those reasons, and the third stage summarizes the reactions to the counterarguments.

Table A1: Summaries for Nuclear Power Focus Group

Discussion Options: 1 = Implement rapid expansion and investment of nuclear power 2 = Maintain the current nuclear energy operations without change 3 = Phase out existing nuclear plants and halt new construction	
Stage 1: Initial Arguments	
Option 1	Nuclear power is viewed as a reliable, low-carbon energy source necessary for meeting climate goals and reducing reliance on fossil fuels. Energy independence and security are seen as benefits of nuclear power, reducing dependence on foreign sources and unreliable renewables. Technological advancements in safety and waste management are believed to mitigate concerns associated with nuclear power. Nuclear power is considered a viable solution for meeting increasing energy demands while providing economic benefits such as job creation. Nuclear power is seen as a practical solution for transitioning away from fossil fuels, especially when considering the limitations of intermittent renewable energy sources.
Option 2	A balanced approach is favored, as it allows for the continued use of a significant energy source without the perceived risks of rapid expansion or the negative impacts of premature phase-out. The status quo is seen as a pragmatic solution, given existing political, social, and environmental complexities, as well as the desire to avoid further polarization while utilizing current infrastructure. Concerns regarding the risks associated with nuclear waste and potential accidents are considered significant enough to warrant caution, emphasizing the need for further advancements in safety and waste management before any expansion is considered. Maintaining current nuclear operations allows for a reliable, low-carbon energy source to be preserved, while further research into safer and more sustainable energy solutions is conducted. The current approach is viewed as a responsible middle ground, balancing energy needs with environmental and safety concerns, and allowing for continued research and public discourse before committing to significant changes.
Option 3	The dangers of long-term nuclear waste storage and the potential for catastrophic accidents are considered unacceptable risks. Investment in renewable energy sources is believed to be a more environmentally responsible and sustainable alternative. The risks associated with nuclear energy are viewed as outweighing any low-carbon benefits. Concerns about the safety and potential environmental impact of nuclear energy are prioritized over its advantages.

Table A1: Summaries for Nuclear Power Focus Group (cont.)

Discussion Options:	
1 = Implement rapid expansion and investment of nuclear power	
2 = Maintain the current nuclear energy operations without change	
3 = Phase out existing nuclear plants and halt new construction	
Stage 2: Counterarguments	
Option 1	Long-term safety and waste disposal concerns are seen as unresolved, with the potential for accidents and mismanagement outweighing the benefits, regardless of technological advancements. The high costs, long construction times, and complex supply chains associated with nuclear power are viewed as potential impediments to a rapid energy transition, diverting resources from more readily available and resilient renewable solutions. The necessity of overhauling the existing energy infrastructure to prioritize nuclear power is questioned, particularly when current needs are being met and other renewable options are available. Prioritizing nuclear power for energy independence is believed to create new vulnerabilities and distract from investments in domestic renewable energy sources. The potential long-term risks associated with rapid nuclear expansion, including waste disposal and safety concerns, are considered to outweigh the economic benefits of job creation and meeting energy demands.
Option 2	A reliance on fossil fuels is seen to be prolonged by not expanding nuclear power, potentially missing critical climate targets due to the limitations of renewable energy sources alone. The risks associated with nuclear power, such as waste disposal and potential accidents, are considered to outweigh the short-term benefits, with a preference for prioritizing a rapid transition to renewable energy sources. A delay in scaling up nuclear power is viewed as a risk to energy security and climate action, as nuclear is currently seen as a readily available solution, and scaling up is preferred while pursuing other innovations. Maintaining the current status of nuclear operations is considered insufficient for achieving necessary emission reductions, and a more proactive approach involving new construction is seen as necessary to displace fossil fuels and meet energy demands. Continuing current nuclear operations is seen as delaying the transition to truly sustainable and secure energy solutions, with concerns that it perpetuates reliance on outdated systems and diverts resources from renewable alternatives.
Option 3	The reliable baseload power provided by nuclear energy is considered essential for grid stability and meeting energy demands, especially when renewables are intermittent or not yet scalable. Modern advancements in nuclear technology and stringent safety protocols are believed to have significantly mitigated risks associated with nuclear energy, making it

Table A1: Summaries for Nuclear Power Focus Group (cont.)

	Discussion Options: 1 = Implement rapid expansion and investment of nuclear power 2 = Maintain the current nuclear energy operations without change 3 = Phase out existing nuclear plants and halt new construction
(cont.)	a safe and manageable energy source. The risks associated with nuclear energy, including waste management and potential accidents, are viewed as manageable with proper technology and regulation, and are outweighed by the existential threat of climate change and the harms of fossil fuels. Nuclear energy is seen as a critical component in transitioning to a low-carbon economy and achieving energy independence, offering a proven, scalable, and low-carbon alternative to fossil fuels without the ecological trade-offs of renewables. Prematurely phasing out existing nuclear plants is viewed as counterproductive, potentially leading to increased reliance on fossil fuels and hindering progress towards climate goals, as well as wasting existing infrastructure.
Stage 3: Responses to Counterarguments	
Option 1	Nuclear power is believed to be a necessary component of a diversified energy portfolio, offering a reliable baseload power source that complements intermittent renewable energy sources and reduces reliance on fossil fuels. Concerns about the safety and waste disposal associated with nuclear power are acknowledged, but it is argued that advancements in reactor designs, waste management technologies, and regulatory oversight have significantly mitigated these risks, making them manageable. The expansion of nuclear power is considered an investment in long-term energy independence, providing a stable and secure energy supply that reduces vulnerability to volatile global markets and geopolitical instability. The urgency of addressing climate change is emphasized, with nuclear power viewed as a crucial tool for decarbonizing the energy sector quickly and effectively, especially given the limitations of renewables in meeting baseload power demands. Modernizing infrastructure and investing in nuclear power is seen as a means of future-proofing the energy grid, preparing for increased energy demands and ensuring a sustainable energy supply for future generations.

Table A1: Summaries for Nuclear Power Focus Group (cont.)

Discussion Options: 1 = Implement rapid expansion and investment of nuclear power 2 = Maintain the current nuclear energy operations without change 3 = Phase out existing nuclear plants and halt new construction	
Stage 3: Responses to Counterarguments (cont.)	
Option 2	A rapid expansion of nuclear power is viewed as unnecessary, as renewable energy sources are believed to have the potential for further development and could meet energy needs. Maintaining current nuclear operations is seen as a pragmatic approach to ensure a stable energy supply while improvements to waste management and safety technologies are made and renewables are scaled up. Decommissioning existing nuclear plants before renewables are fully capable of meeting demand is viewed as risky, potentially creating energy instability and increasing reliance on fossil fuels. Building new nuclear plants is seen as a potentially risky and costly endeavor, with concerns raised about waste disposal, long-term risks, and the prioritization of renewable energy sources. A cautious approach to nuclear energy is favored, focusing on optimizing existing plants, improving safety, and exploring the potential of renewable energy sources before committing to large-scale new construction.
Option 3	The long-term risks associated with nuclear power, such as waste disposal and potential accidents, are believed to outweigh the benefits, especially when renewable energy sources are available. Reliance on nuclear power is viewed as unnecessary due to the availability of cleaner, safer, and more sustainable alternatives like solar and wind power. Concerns are raised that the term manageable does not equate to safe, particularly in the context of nuclear power's potential for catastrophic accidents and long-term environmental impact. The possibility of a short-term energy gap resulting from phasing out nuclear plants is acknowledged, but it is suggested that this gap can be bridged with investments in renewable energy and storage solutions. Continued investment in nuclear power is seen as diverting resources from the development and deployment of renewable energy technologies.